

Introduction to Big Data, Big Data Processing, and Cloud Computing Systems for Big Data

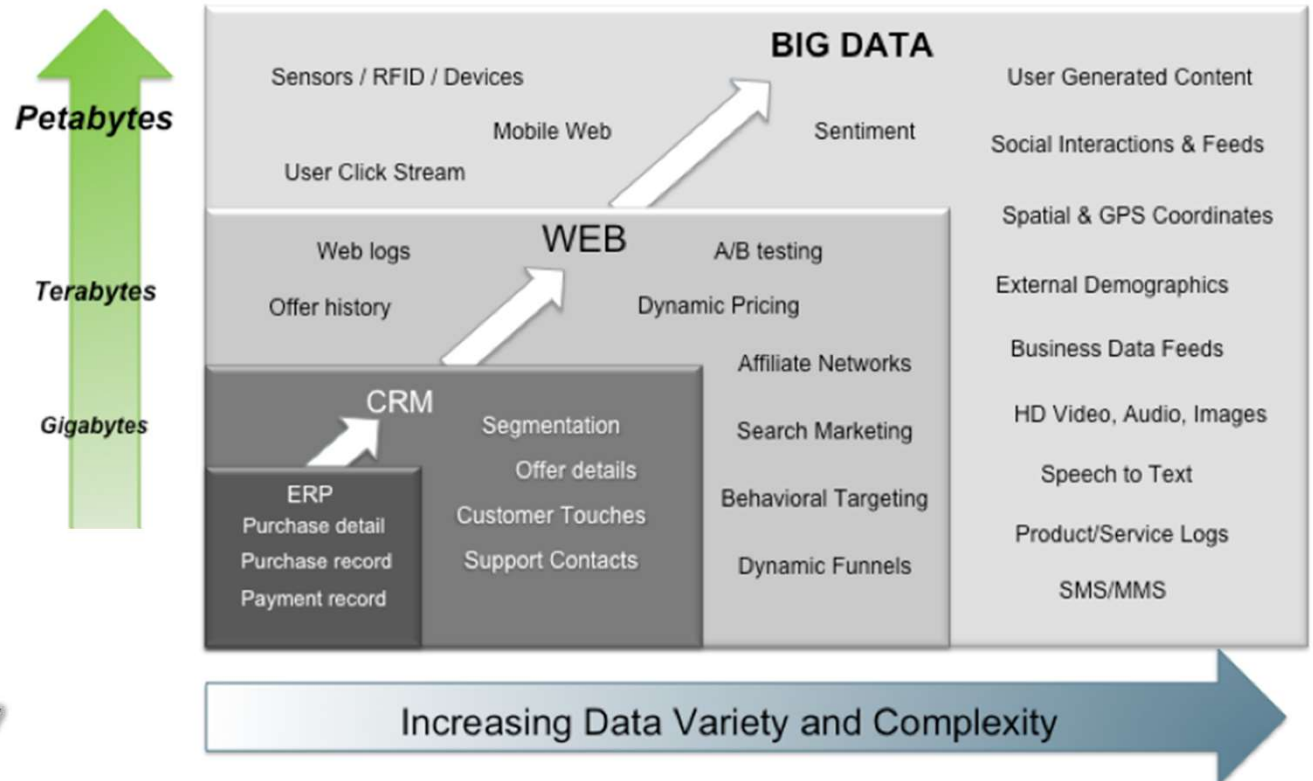
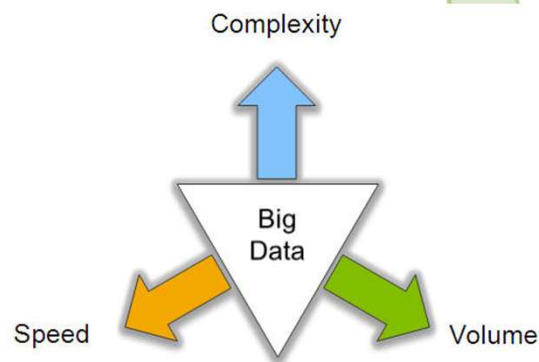
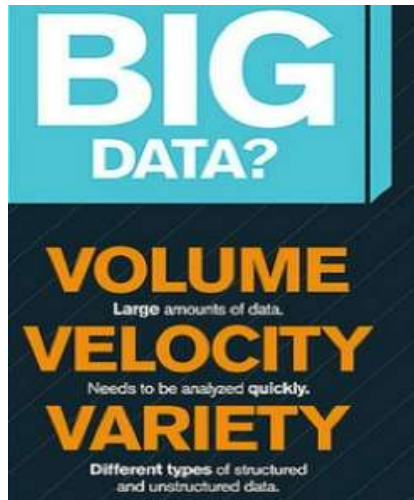
What's Big Data?

From Wikipedia:

- **Big data** is the term for a collection of data sets so large and complex that it becomes difficult to process using on-hand database management tools or traditional data processing applications.
- The challenges include **capture, curation, storage, search, sharing, transfer, analysis, and visualization**.
- The trend to larger data sets is due to the additional information derivable from analysis of a single large set of related data, as compared to separate smaller sets with the same total amount of data, allowing correlations to be found to "**spot business trends, determine quality of research, prevent diseases, link legal citations, combat crime, and determine real-time roadway traffic conditions.**"

Big Data: 3V's

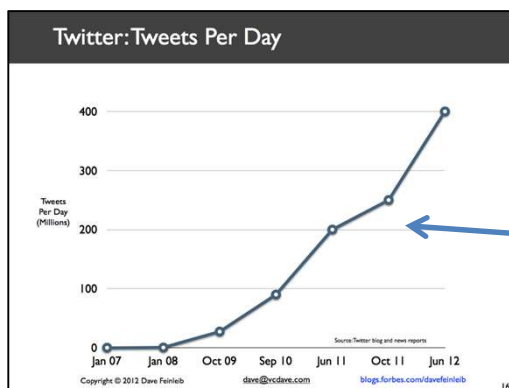
Big Data = Transactions + Interactions + Observations



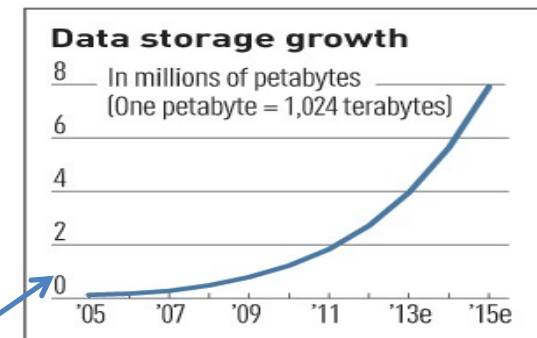
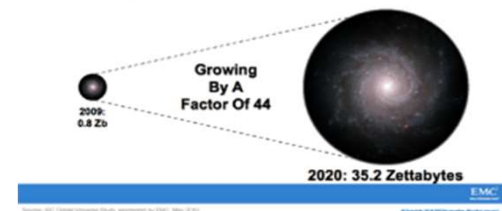
Source: Contents of above graphic created in partnership with Teradata, Inc.

Volume (Scale)

- **Data Volume**
 - 44x increase from 2009 to 2022
 - From 0.8 zettabytes to 35zb
- Data volume is increasing exponentially

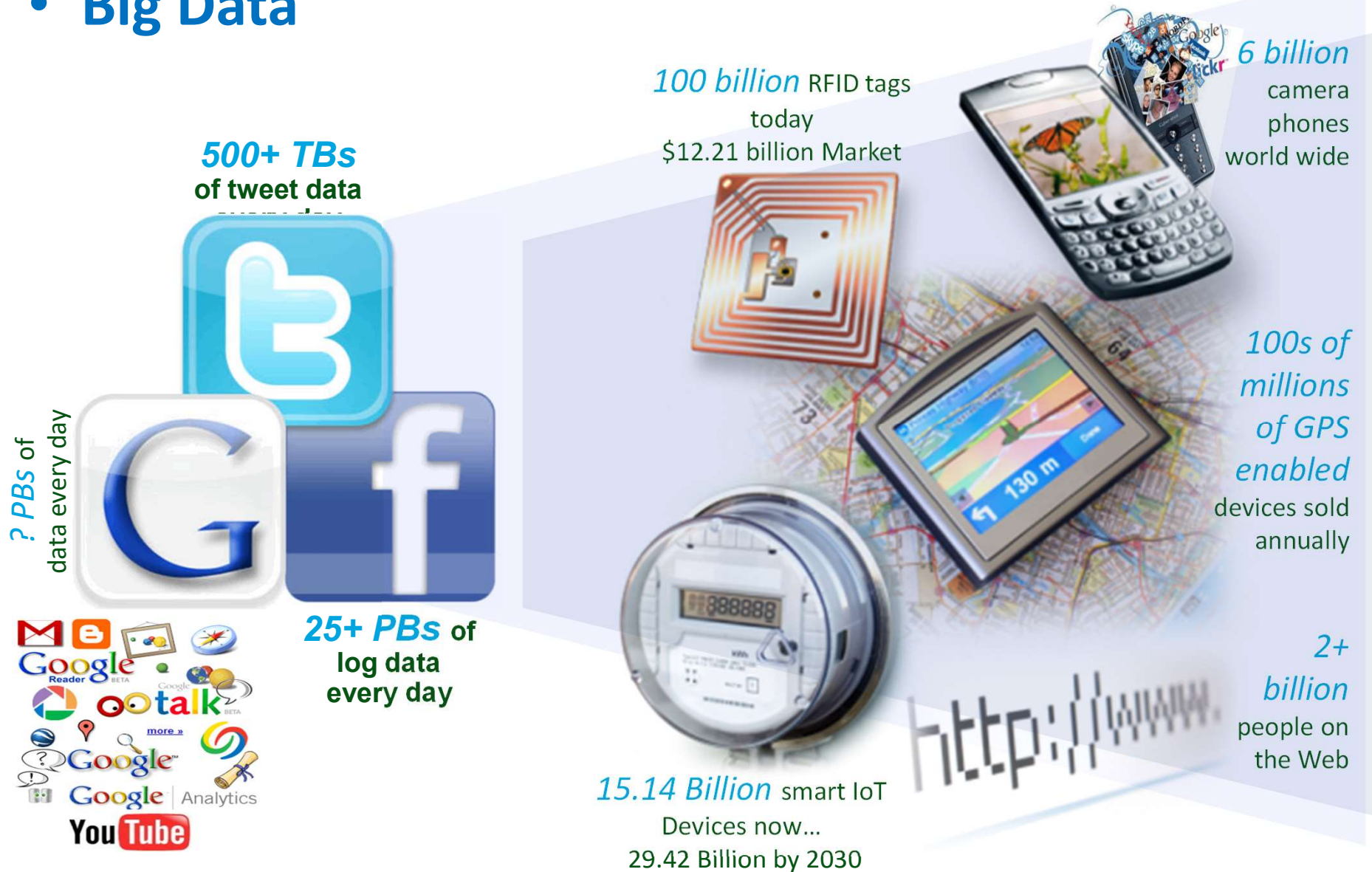


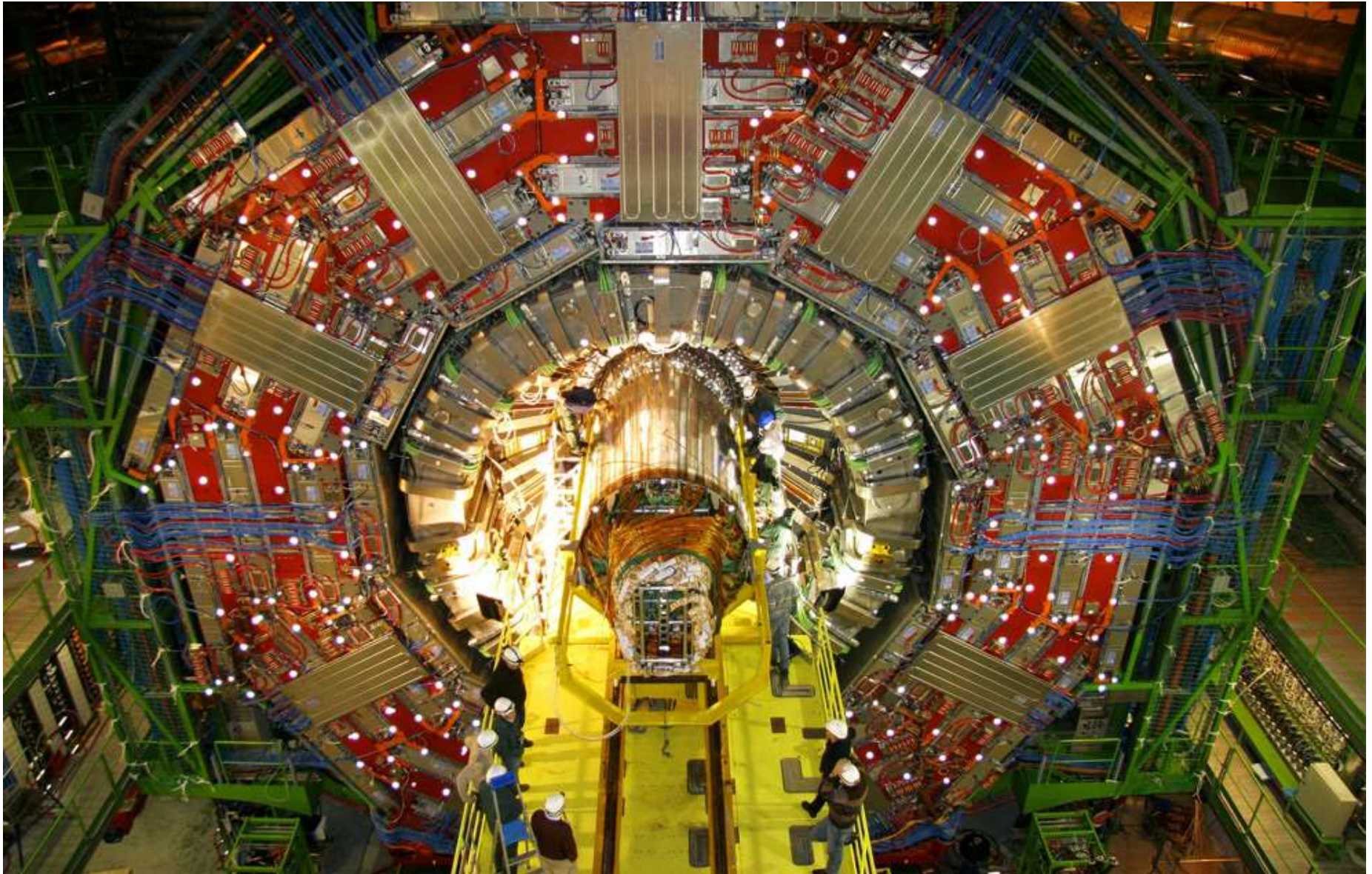
The Digital Universe 2009-2020



Exponential increase in collected/generated data

• Big Data





CERN's Large Hydron Collider (LHC) generates 15 PB a year

Sunnie Chung Cleveland State University

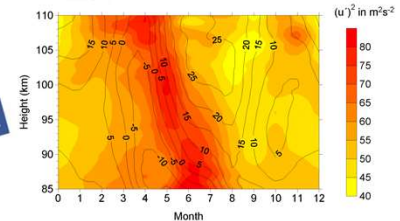
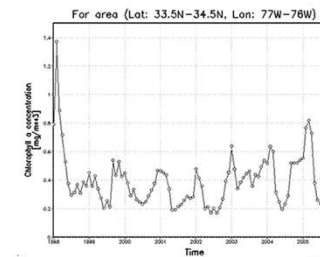
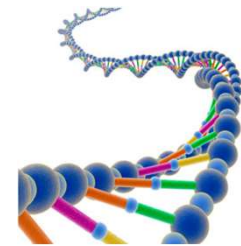
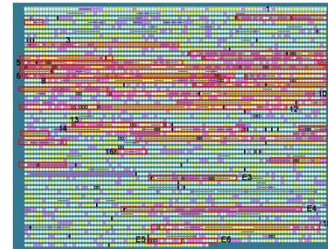
The EarthScope

- The Earthscope is the world's largest science project. Designed to track North America's geological evolution, this observatory records data over 3.8 million square miles, amassing 67 terabytes of data. It analyzes seismic slips in the San Andreas fault, sure, but also the plume of magma underneath Yellowstone and much, much more.
- http://www.msnbc.msn.com/id/44363598/ns/technology_and_science-future_of_technology/#.TmetOdQ--ul



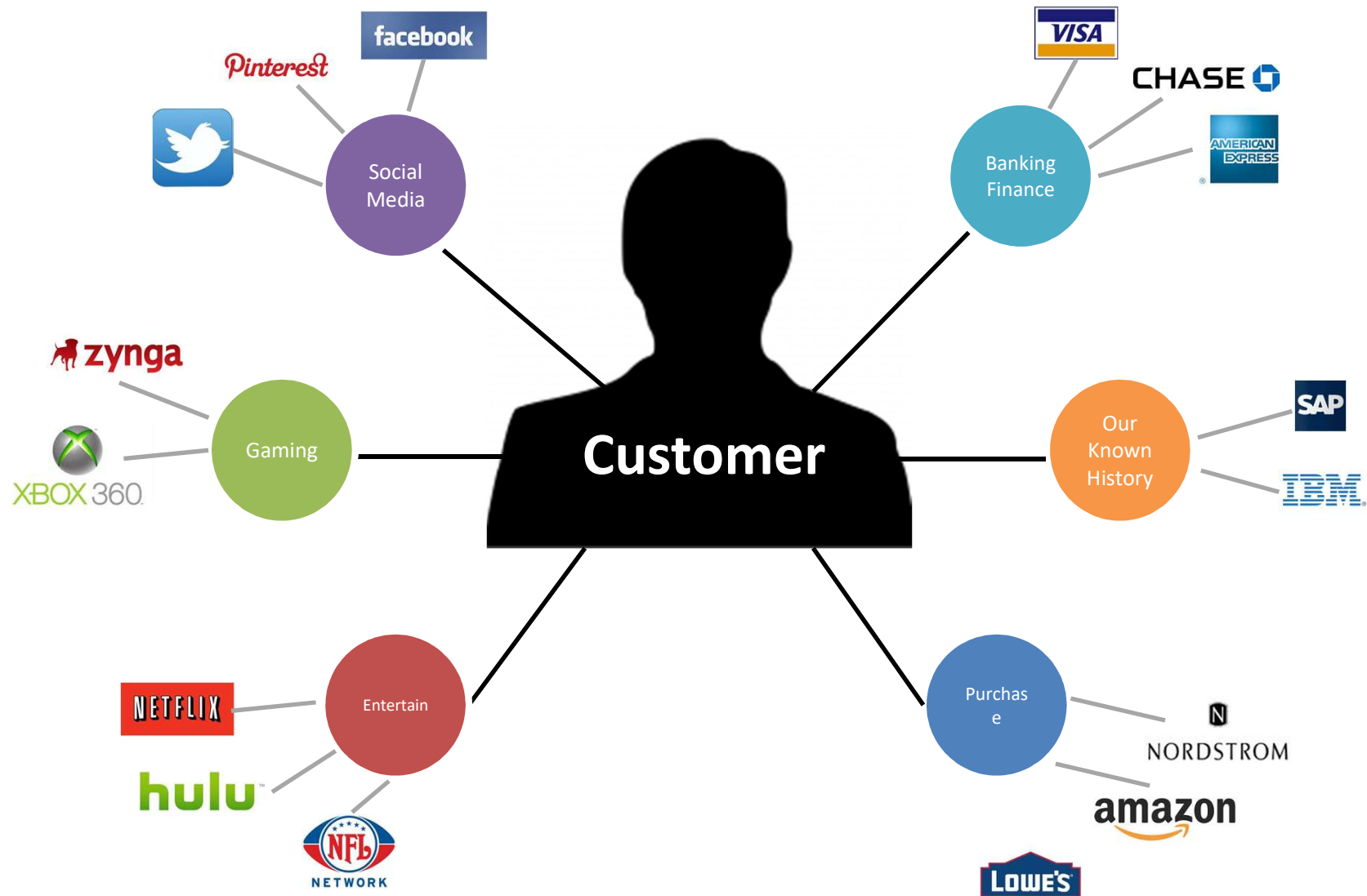
Variety (Complexity)

- Relational Data (Tables/Transaction/Legacy Data)
- Text Data (Web)
- Semi-structured Data (XML)
- Graph Data
 - Social Network, Semantic Web (RDF), ...
- Streaming Data
 - You can only scan the data once
- A single application can be generating/collecting many types of data
- Big Public Data (online, weather, finance, etc)



To extract knowledge → all these types of data need to be linked together

A Single View to the Customer



Velocity (Speed)

- Data is begin generated fast and need to be processed fast
- Online Data Analytics
- Late decisions → missing opportunities



- **Examples**

- **E-Promotions:** Based on your current location, your purchase history, what you like → send promotions right now for store next to you
- **Healthcare monitoring:** sensors monitoring your activities and body → any abnormal measurements require immediate reaction

Real-time/Fast Data



Social media and networks
(all of us are generating data)



Scientific instruments
(collecting all sorts of data)



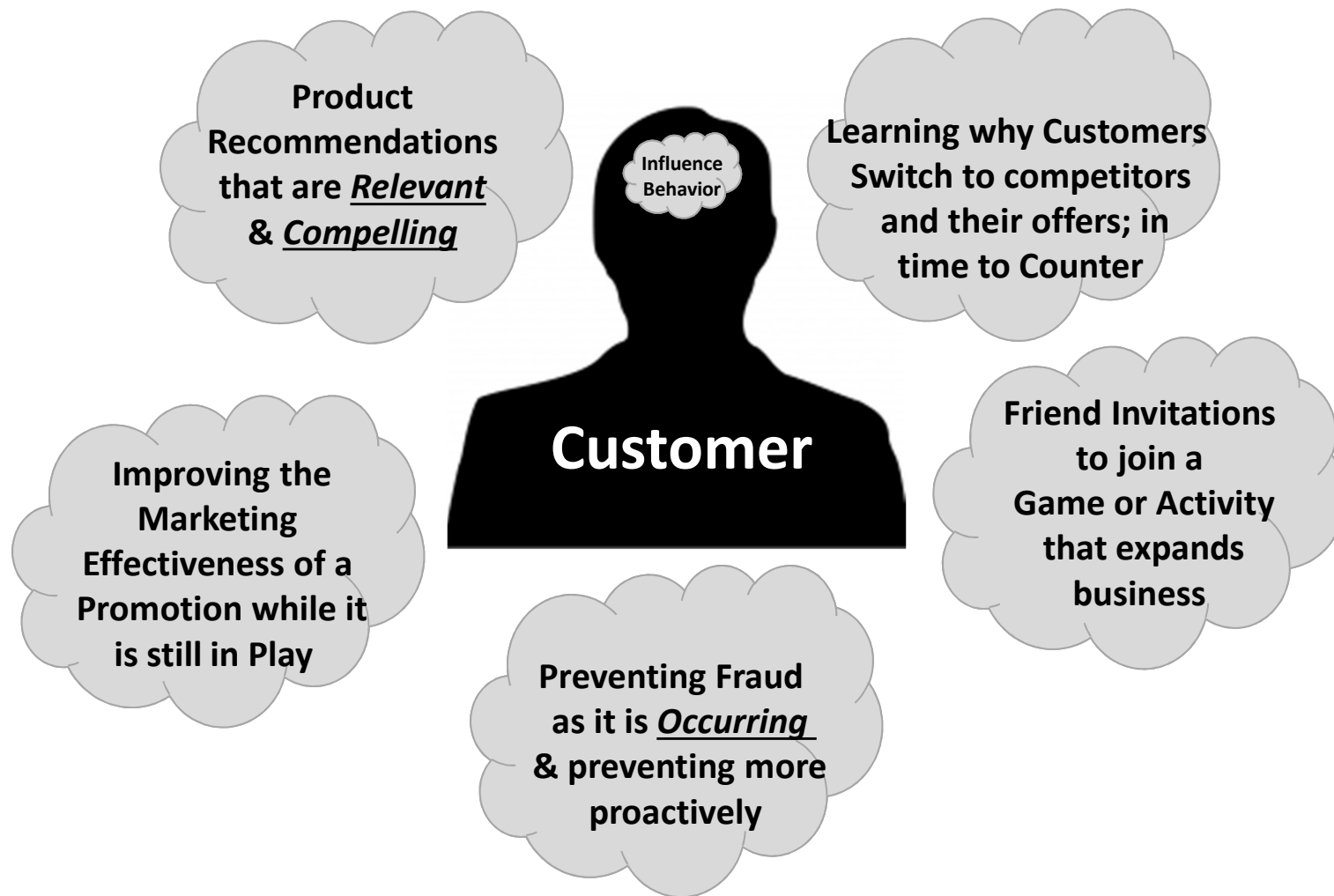
Mobile devices
(tracking all objects all the time)



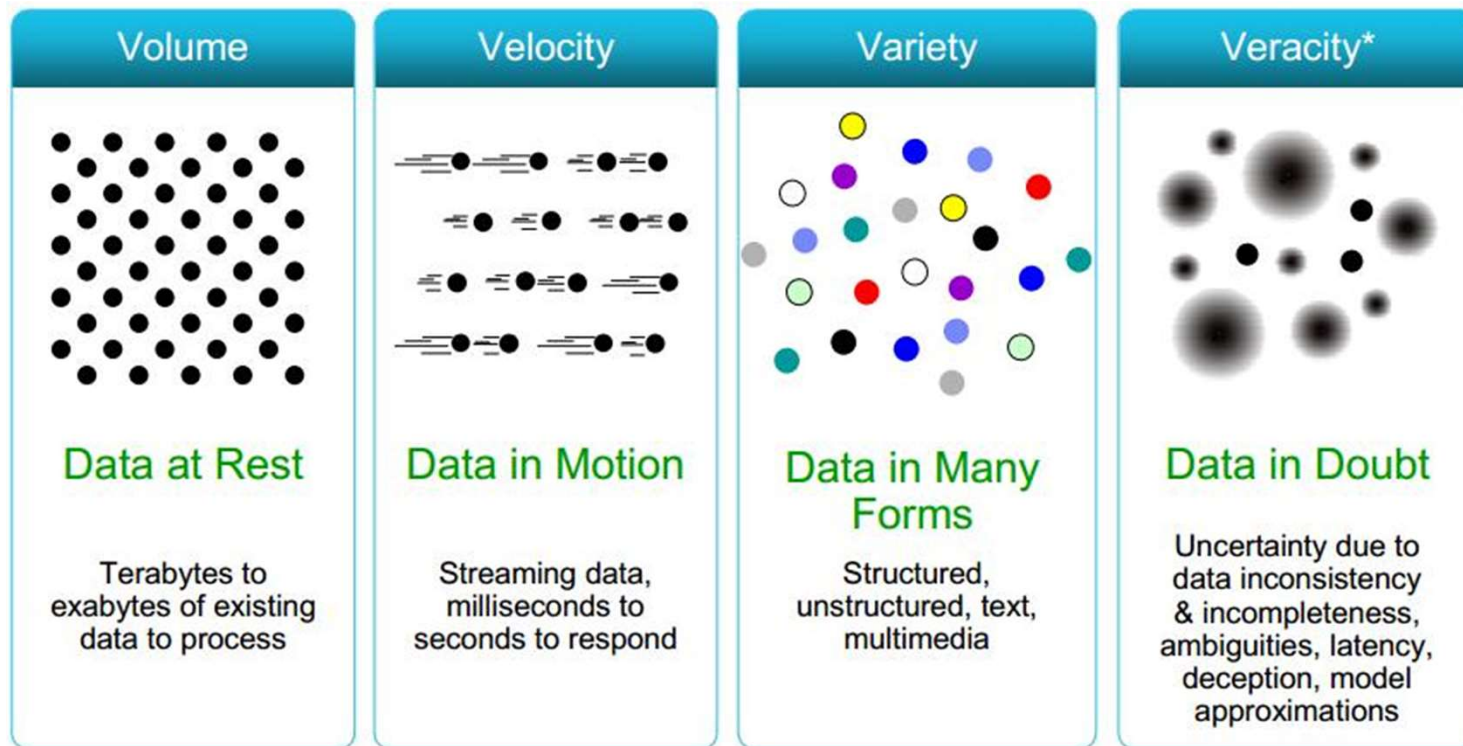
Sensor technology and networks
(measuring all kinds of data)

- The progress and innovation is no longer hindered by the ability to collect data
- But, by the ability to manage, analyze, summarize, visualize, and discover knowledge from the collected data in a timely manner and in a scalable fashion

Real-Time Analytics/Decision Requirement



Some Make it 4V's



The Model Has Changed...

- **The Model of Generating/Consuming Data has Changed**

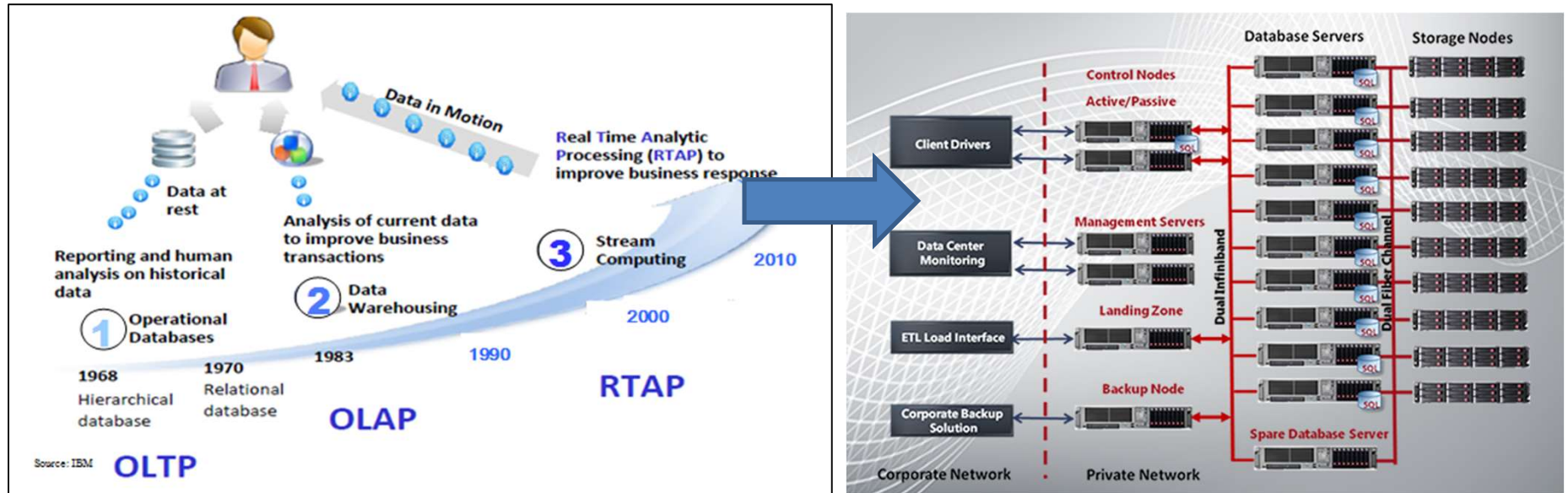
Old Model: Few companies are generating data, all others are consuming data



New Model: all of us are generating data, and all of us are consuming data

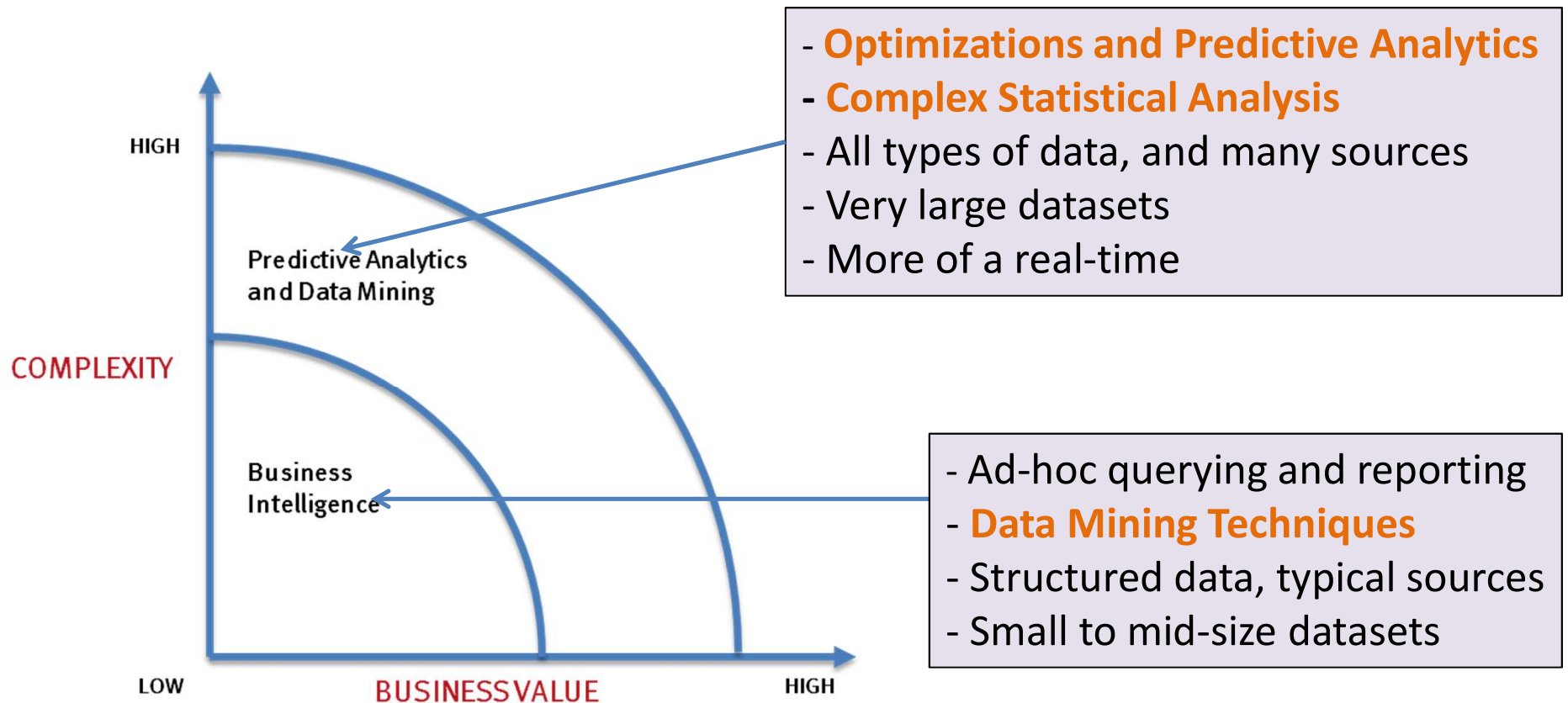


Harnessing Big Data

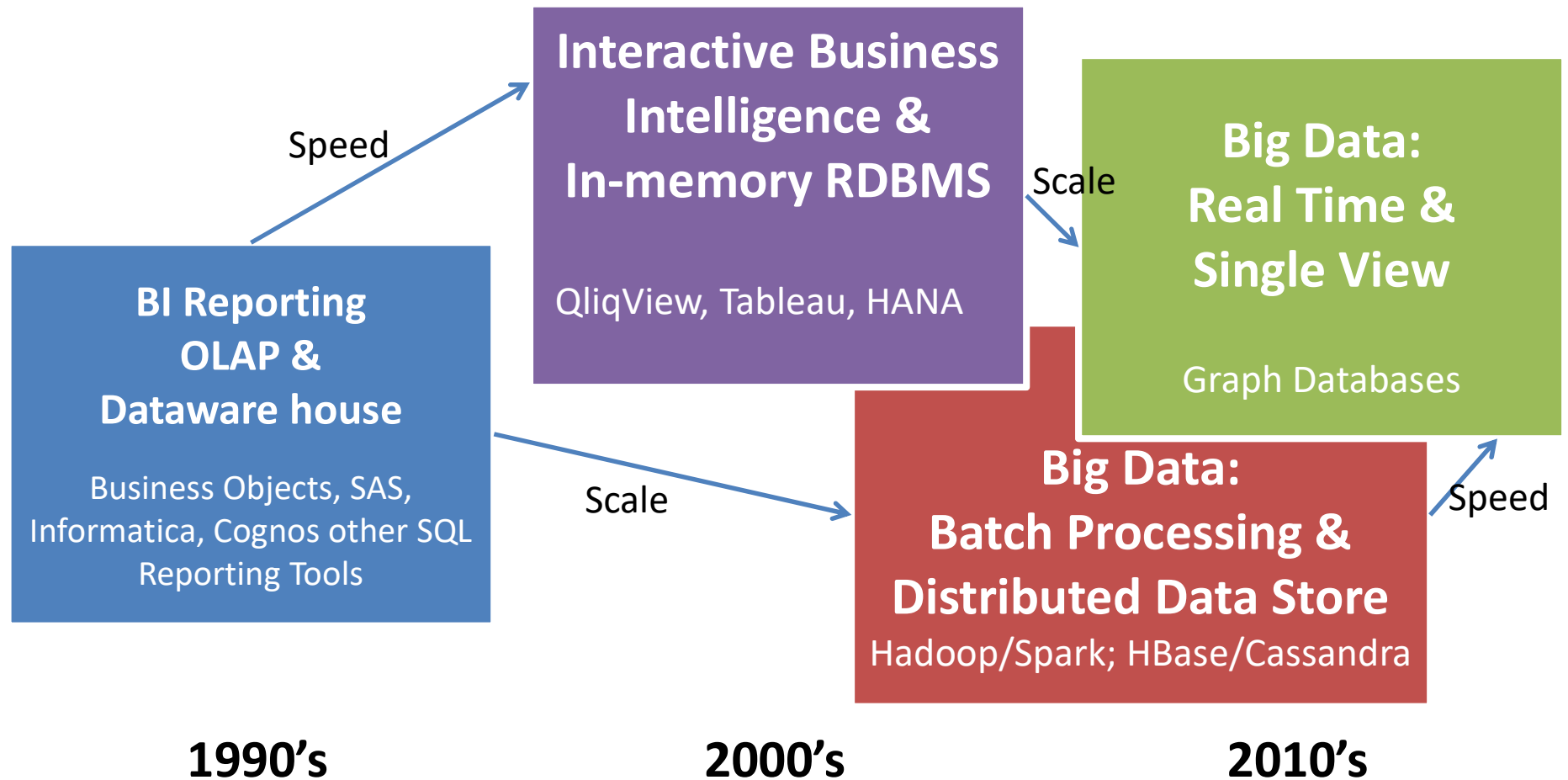


- **OLTP:** Online Transaction Processing (DBMSs)
- **OLAP:** Online Analytical Processing (Data Warehousing) => **Parallel DW (PDW)**
- **RTAP:** Real-Time Analytics Processing (Big Data Architecture & Technology)

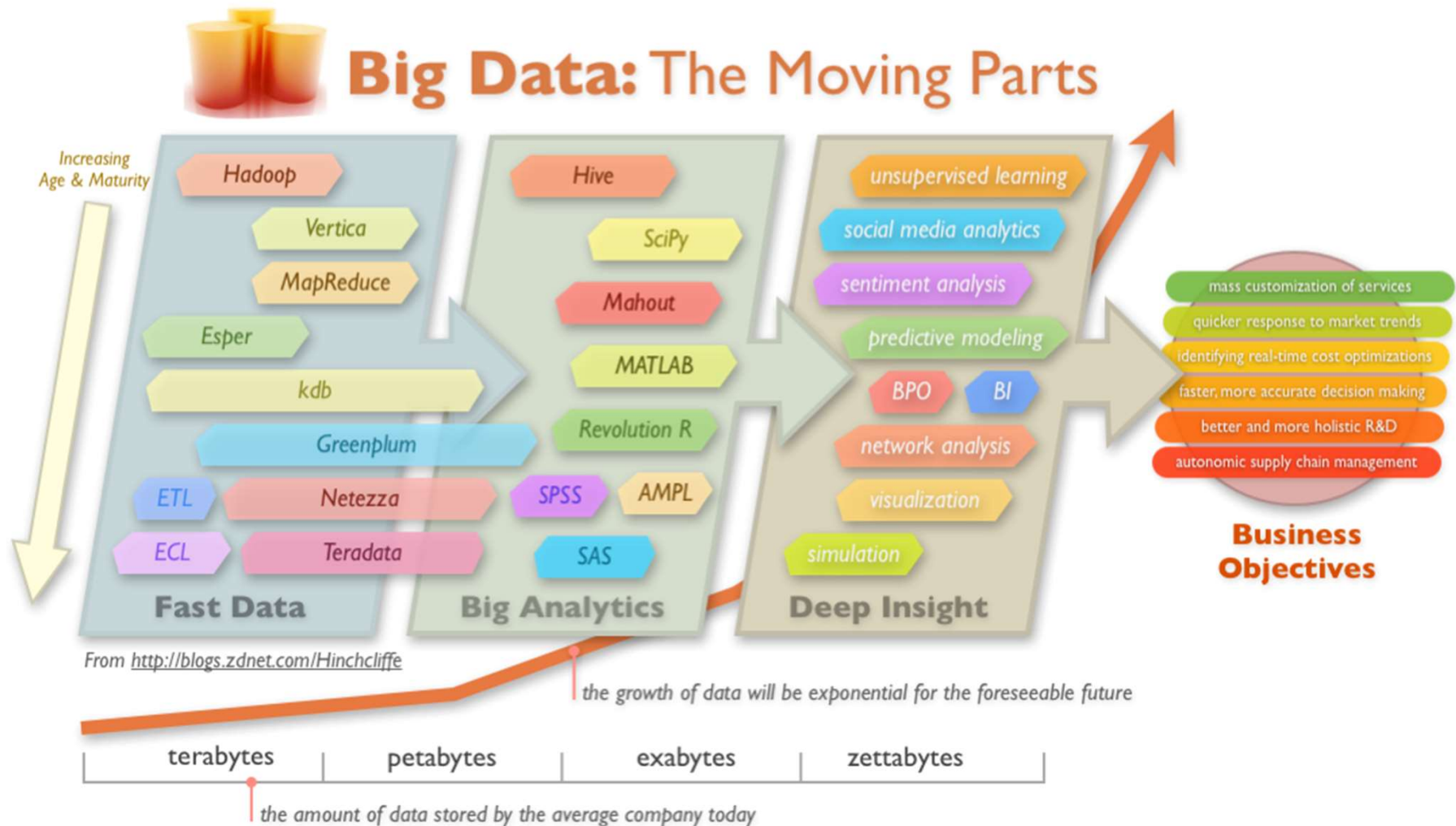
What's Driving Big Data



THE EVOLUTION OF BUSINESS INTELLIGENCE



Big Data Technology

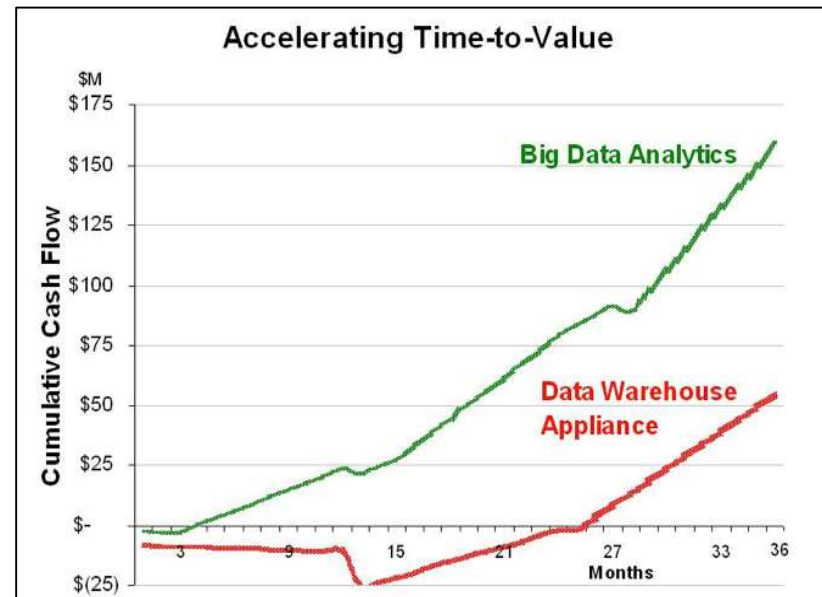


Big Data Analytics

- Big data is more real-time in nature than traditional DW applications
- Traditional DW architectures (e.g. Oracle, IBM) are not well-suited for big data apps
- **Shared Nothing, Massively Parallel Processing (MPP)**, scale out architectures are well-suited for big data apps

MPP Industry:

- **AMAZON Elastic Cloud Computing (EC2) Cluster**
- **AZURE Parallel Data Warehouse**



The Big Data Landscape

Apps

Vertical Apps



Operational Intelligence



Data As A Service



Ad / Media Apps



Business Intelligence



Analytics And Visualization



Infrastructure

Analytics Infrastructure



Operational Infrastructure



Infrastructure As A Service



Structured Databases



Technologies

Main Big Data Technologies

Hadoop

- Low cost, reliable scale-out architecture
- Distributed computing
Proven success in Fortune 500 companies
- Exploding interest



Hadoop

NoSQL Databases

- Huge horizontal scaling and high availability
- Highly optimized for retrieval and appending
- Types
 - Document stores
 - Key Value stores
 - Graph databases



NoSQL Databases

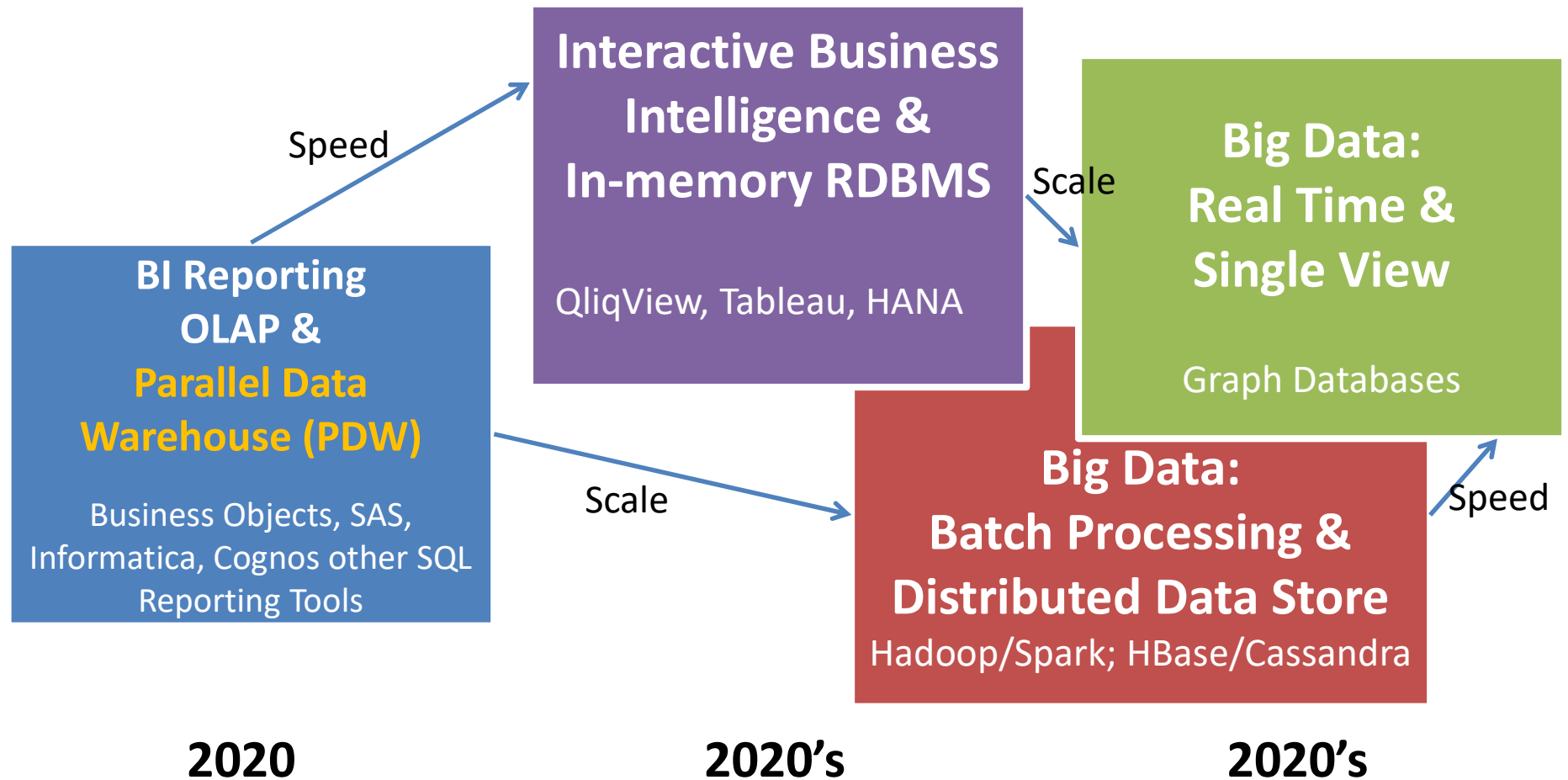
Analytic RDBMS

- Optimized for bulk-load and fast aggregate query workloads
- Types
 - Column-oriented
 - MPP
 - In-memory



Analytic Databases

CURRENT ENTERPRISE BIG DATA PROCESSING SYSTEMS FOR BUSINESS INTELLIGENCE: ALL THREE TOGETHER



Big Data Analytics

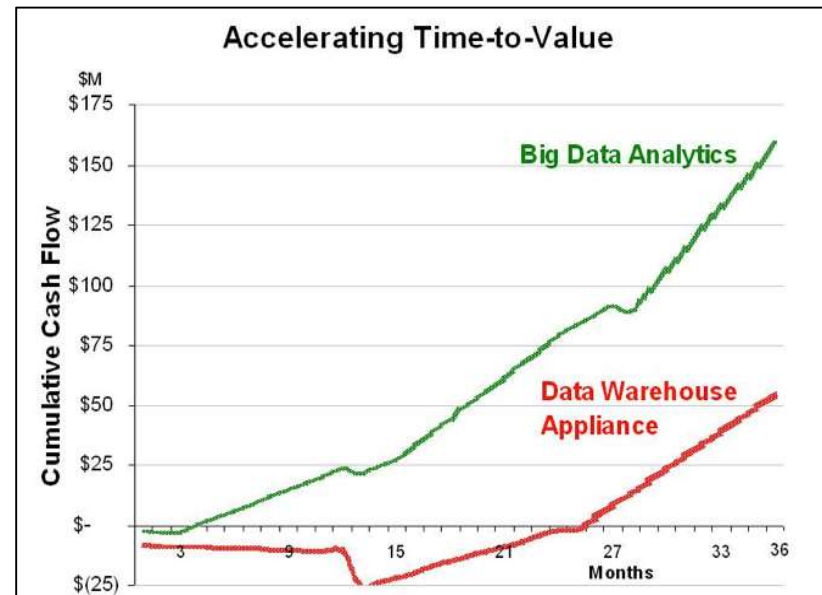
- Big data is more real-time in nature than traditional DW applications
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➔ Optimized DW Systems:

- InMemory DB, Columnar DB (SAP HANA)
- Shared Nothing, Massively Parallel Processing scale out architectures are well-suited for big data apps

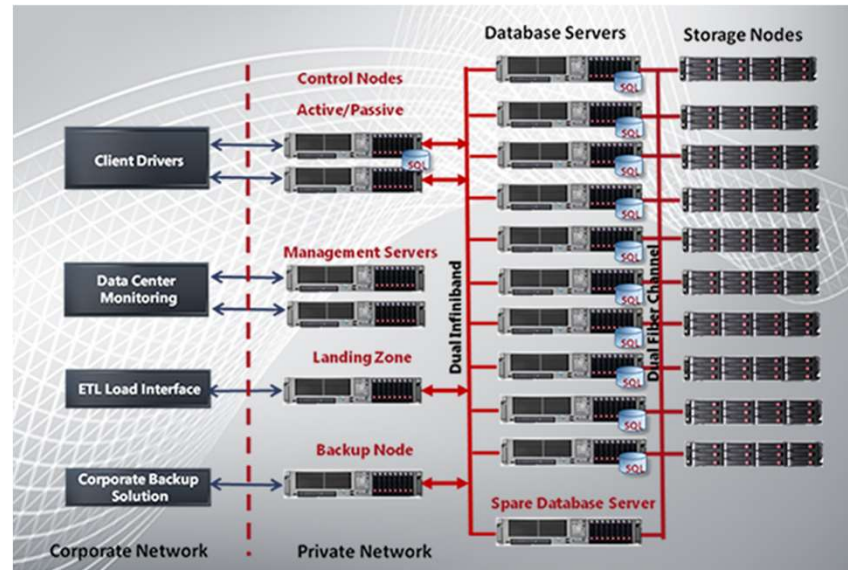
Industry Examples:

- Microsoft Parallel DW (PDW)
- Tera Data MPP PDW



Shared Nothing, Massively Parallel Processing (MPP) Systems

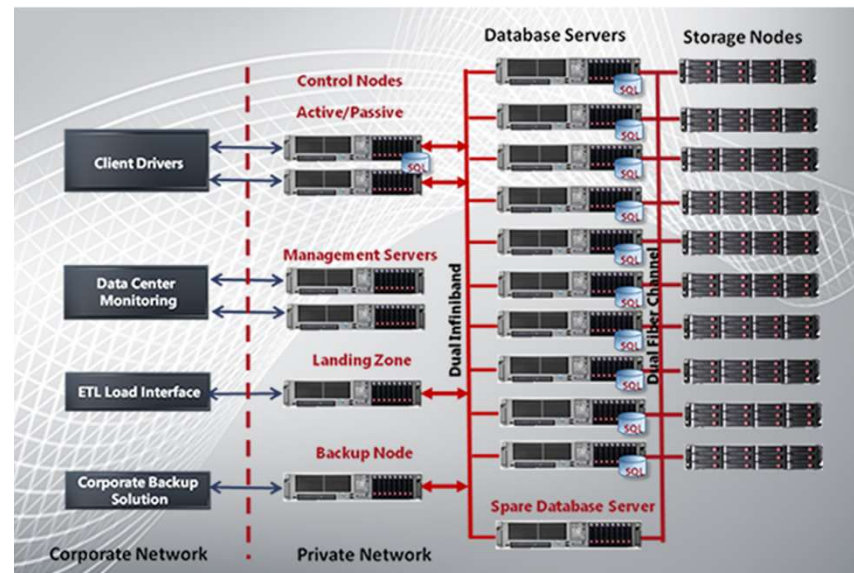
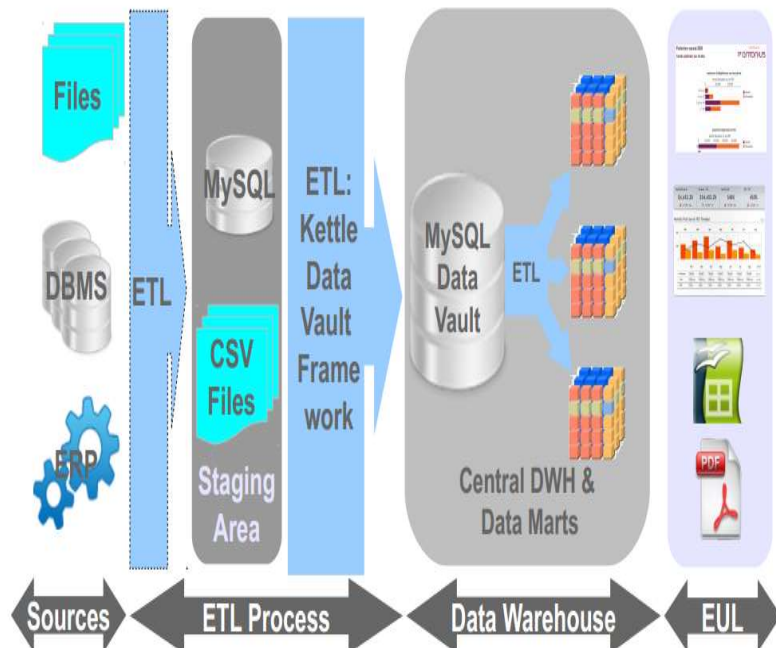
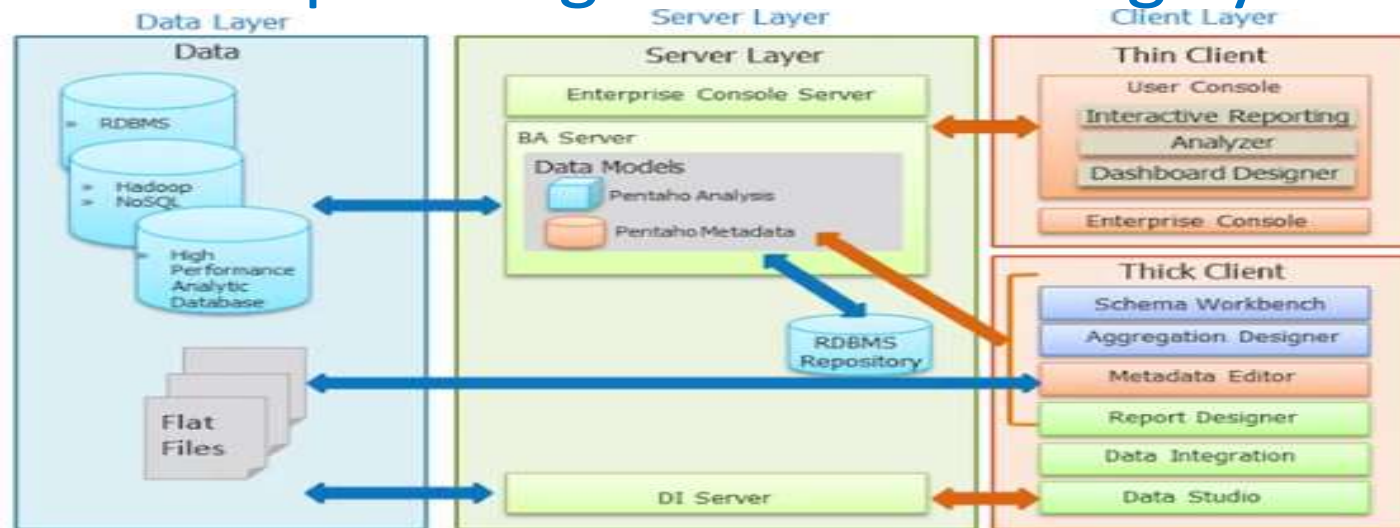
Shared Nothing, Massively Parallel Processing (MPP), scale out architectures are well-suited for big data apps



MPP Industry:

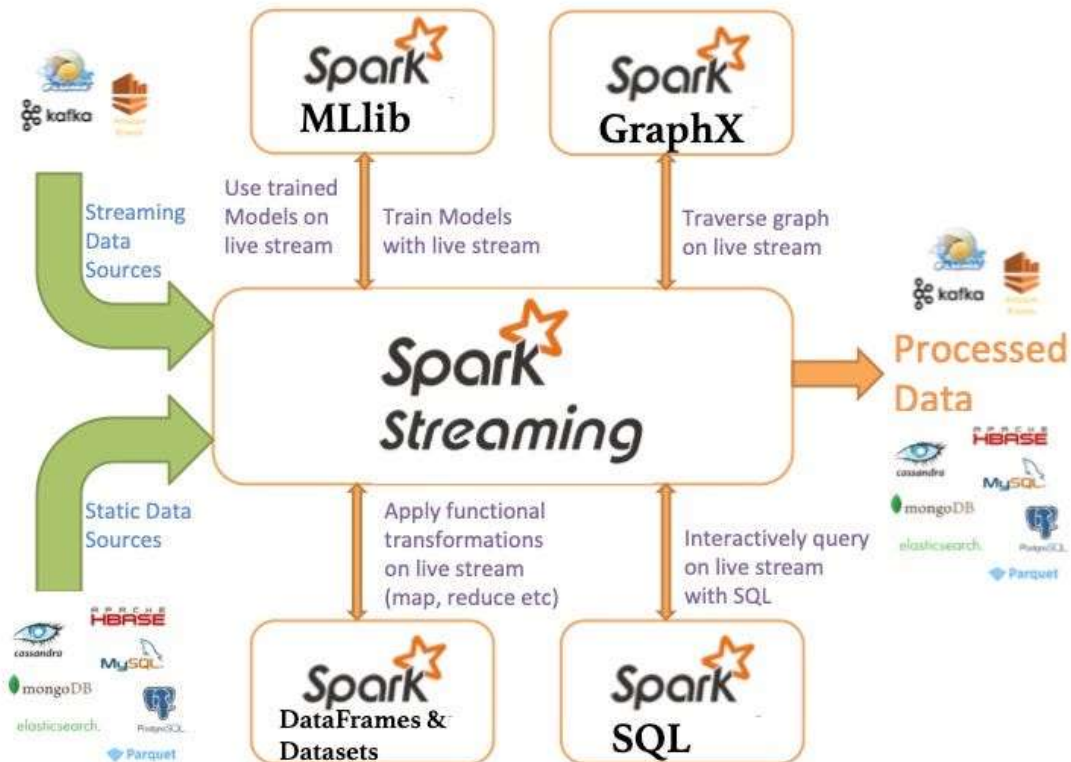
- AMAZON Elastic Cloud Computing (EC2) Cluster
- AZURE Parallel Data Warehouse

Enterprise Big Data Processing Systems



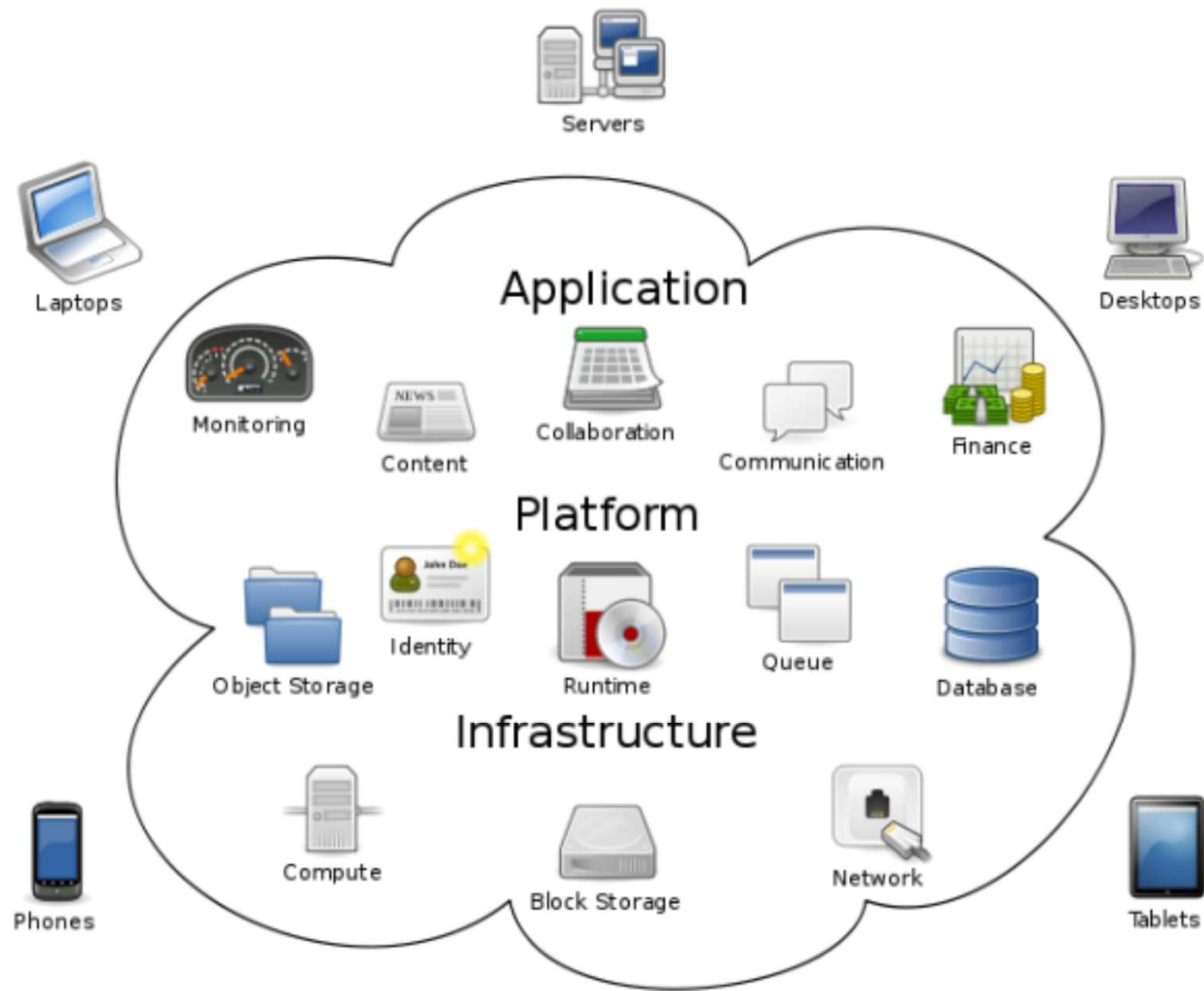
Another New Platform for Big Data Analytics

- In Memory Based Systems
- Spark



Cloud Computing

- IT resources provided as a service
 - Compute, storage, databases, queues
- Clouds leverage economies of scale of commodity hardware
 - Cheap storage, high bandwidth networks & multicore processors
 - Geographically distributed data centers
- Offerings from Microsoft, Amazon, Google, ...



Cloud Computing

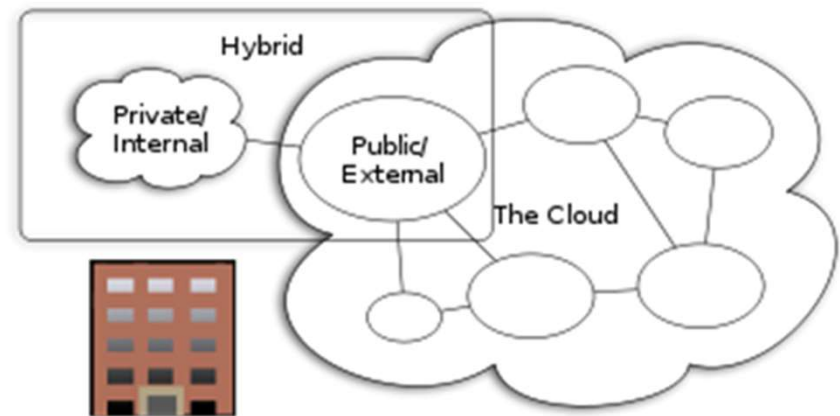
wikipedia:Cloud Computing

Benefits

- Cost & management
 - Economies of scale, “out-sourced” resource management
- Reduced Time to deployment
 - Ease of assembly, works “out of the box”
- Scaling
 - On demand provisioning, co-locate data and compute
- Reliability
 - Massive, redundant, shared resources
- Sustainability
 - Hardware not owned

Types of Cloud Computing

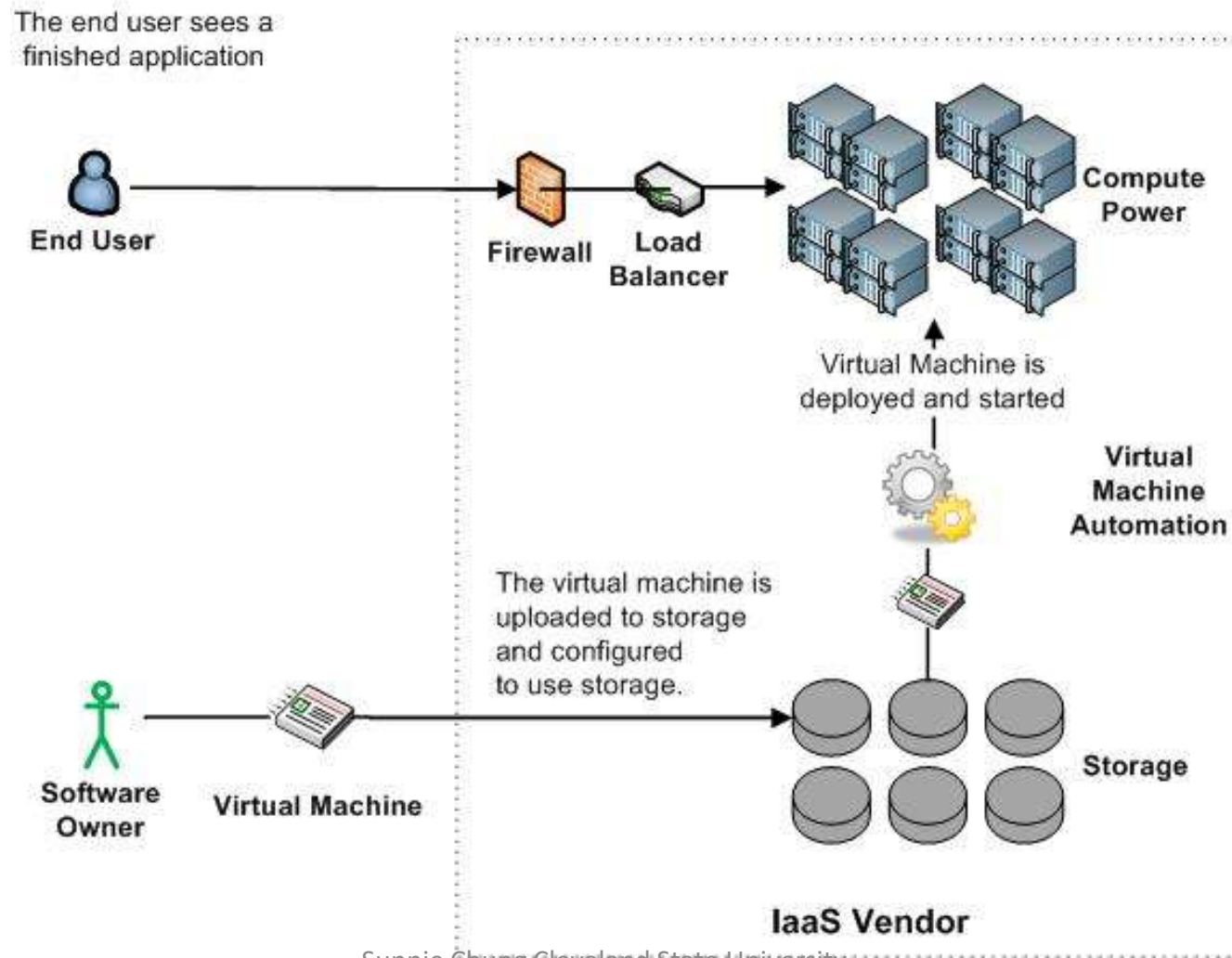
- **Public Cloud:** Computing infrastructure is hosted at the vendor's premises.
- **Private Cloud:** Computing architecture is dedicated to the customer and is not shared with other organisations.
- **Hybrid Cloud:** Organisations host some critical, secure applications in private clouds. The not so critical applications are hosted in the public cloud
 - **Cloud bursting:** the organisation uses its own infrastructure for normal usage, but cloud is used for peak loads.
- **Community Cloud**



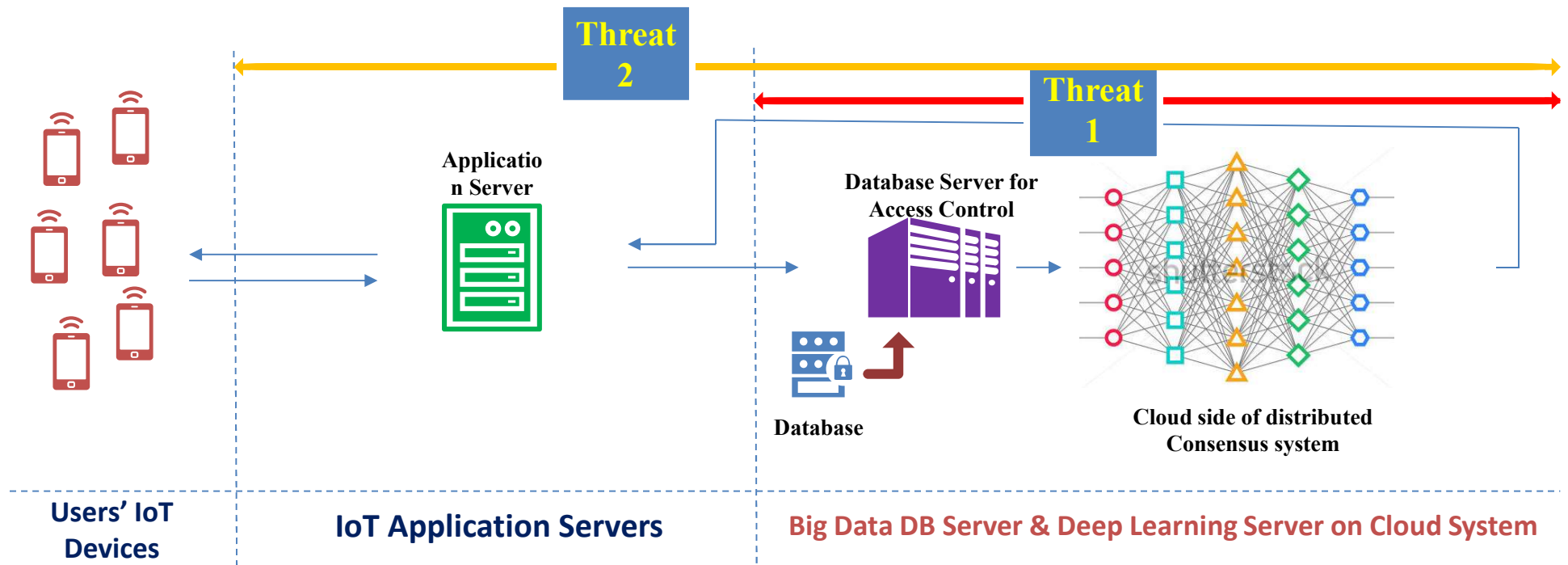
Classification of Cloud Computing based on Service Provided

- Infrastructure as a service (IaaS)
 - Offering hardware related services using the principles of cloud computing. These could include storage services (database or disk storage) or virtual servers.
 - [Amazon EC2](#), [Amazon S3](#), [Rackspace Cloud Servers](#) and [Flexiscale](#).
- Platform as a Service (PaaS)
 - Offering a development platform on the cloud.
 - [Google's Application Engine](#), [Microsofts Azure](#), Salesforce.com's [force.com](#).
- Software as a service (SaaS)
 - Including a complete software offering on the cloud. Users can access a software application hosted by the cloud vendor on pay-per-use basis. This is a well-established sector.
 - Salesforce.com's offering in the online Customer Relationship Management (CRM) space, Google's [gmail](#) and Microsoft's [hotmail](#), [Google docs](#).

Infrastructure as a Service (IaaS)



Architecture of Cloud Based IoT Applications



More Refined Categorization

- Storage-as-a-service
- **Database-as-a-service**
- **Information-as-a-service**
- Process-as-a-service
- Application-as-a-service
- **Platform-as-a-service**
- Integration-as-a-service
- **Security-as-a-service**
- Management/
Governance-as-a-service
- Testing-as-a-service
- Infrastructure-as-a-service
- **Machine Learning-as-service**

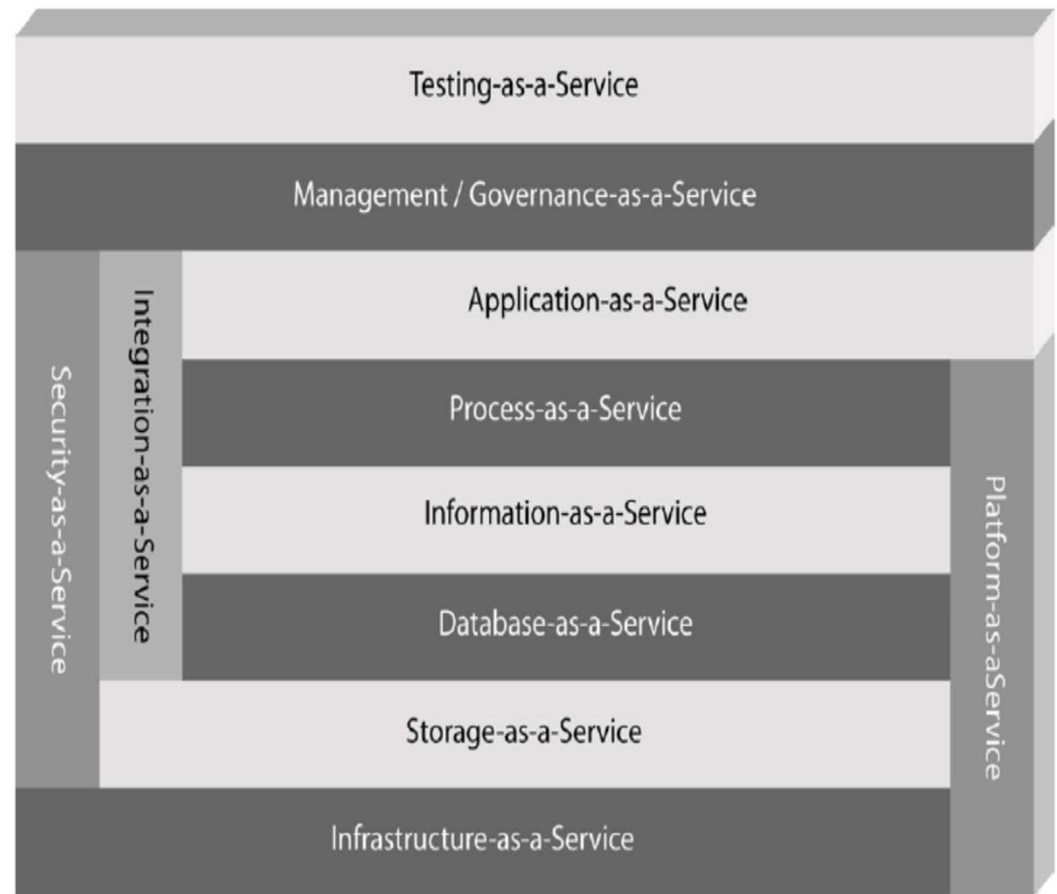


Figure 1: The patterns or categories of cloud computing providers allow you to use a discrete set of services within your architecture.

InfoWorld Cloud Computing Deep Dive

Key Ingredients in Cloud Computing

- Service-Oriented Architecture (SOA)
- Utility Computing (on demand)
- Virtualization (P2P Network)
- SAAS (Software As A Service)
- PAAS (Platform AS A Service)
- IAAS (Infrastructure AS A Servie)
- Web Services in Cloud

Everything as a Service

- Utility computing = Infrastructure as a Service (IaaS)
 - Why buy machines when you can rent cycles?
 - Examples: Amazon's EC2, Rackspace
- Platform as a Service (PaaS)
 - Give me nice API and take care of the maintenance, upgrades, ...
 - Example: Google App Engine
- Software as a Service (SaaS)
 - Just run it for me!
 - Example: Gmail, Salesforce

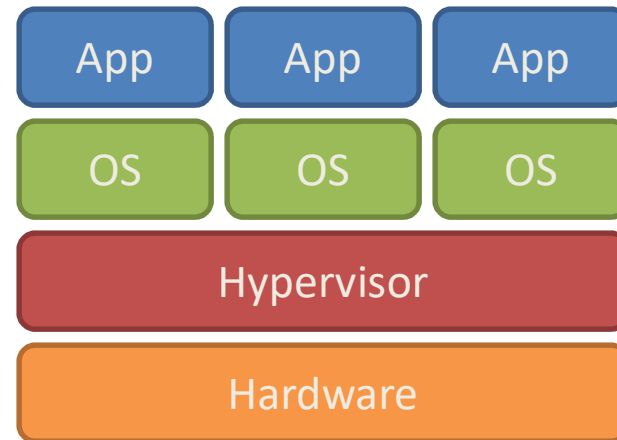
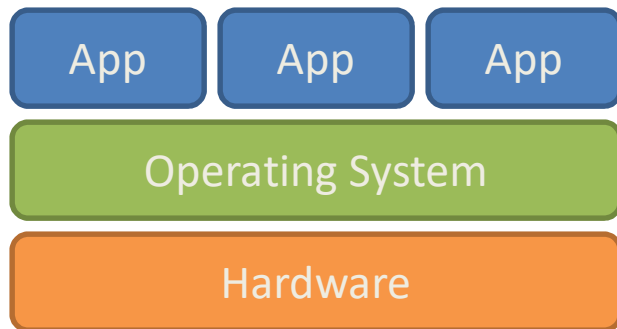
Cloud Computing Systems

- Amazon Elastic Compute Cloud
- Google App Engine
- Microsoft Azure
- GoGrid
- AppNexus

Amazon Web Service (AWS)

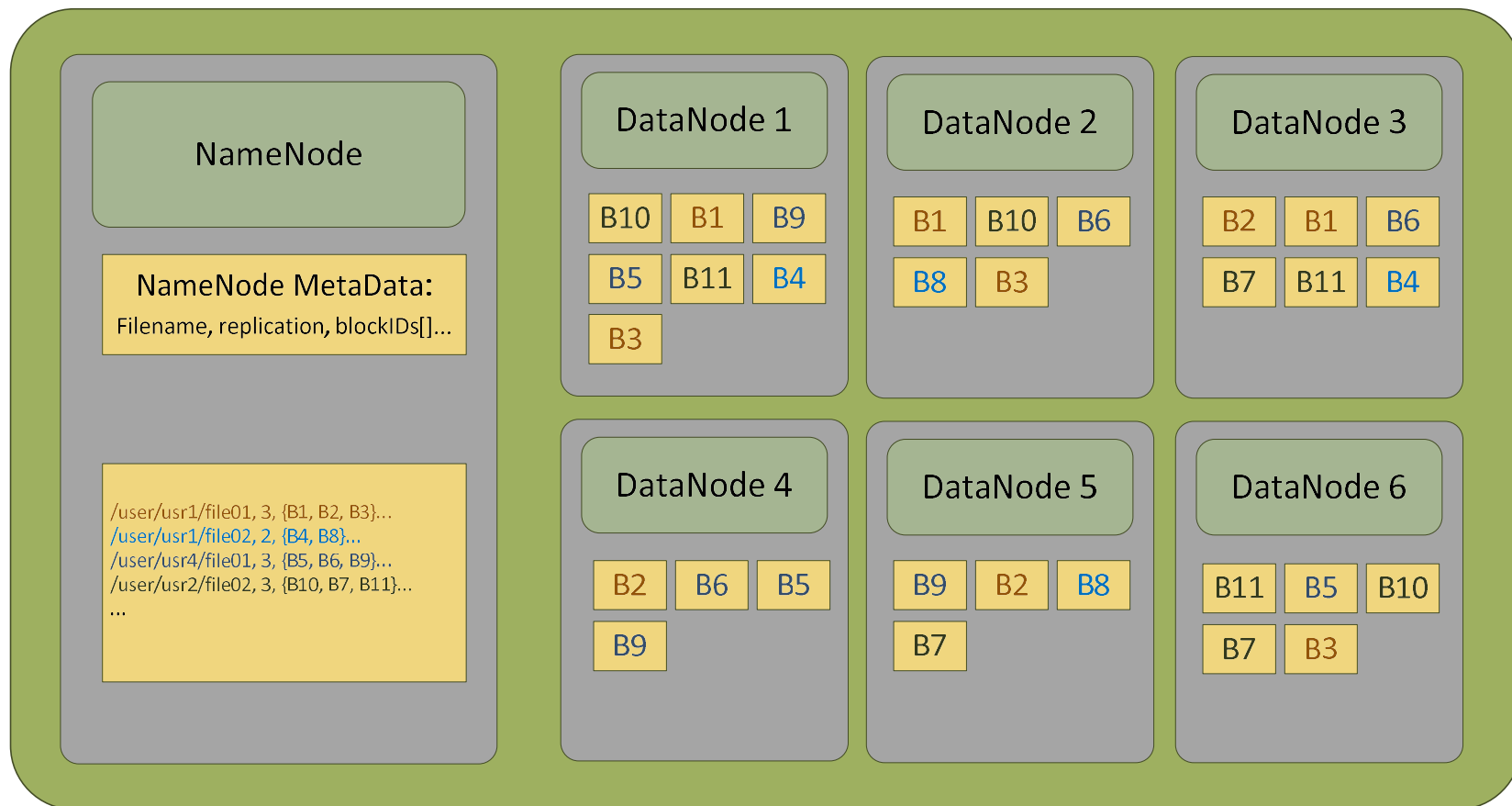
- Elastic Compute Cloud – EC2 (IaaS)
- Simple Storage Service – S3 (IaaS)
- Elastic Block Storage – EBS (IaaS)
- SimpleDB (SDB) (PaaS)
- Simple Queue Service – SQS (PaaS)
- CloudFront (S3 based Content Delivery Network – PaaS)
- Consistent AWS Web Services API

Enabling Technology: **Virtualization**

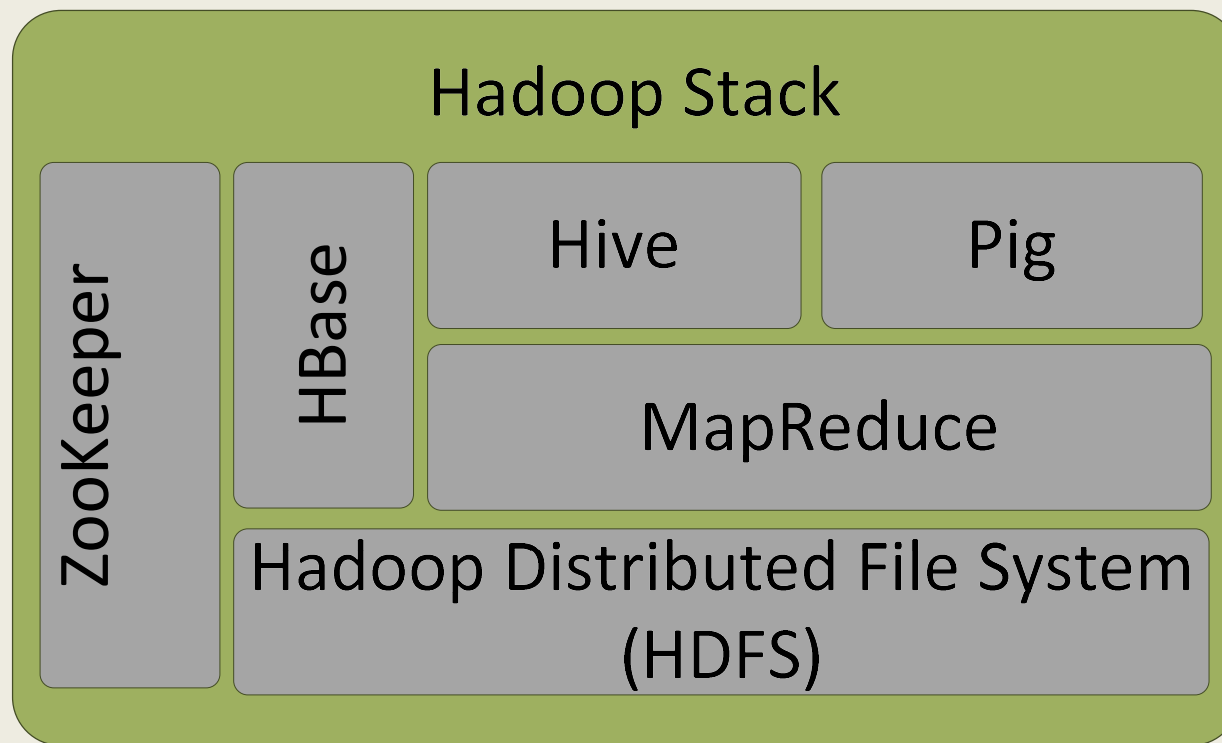


Parallel Distributed File System

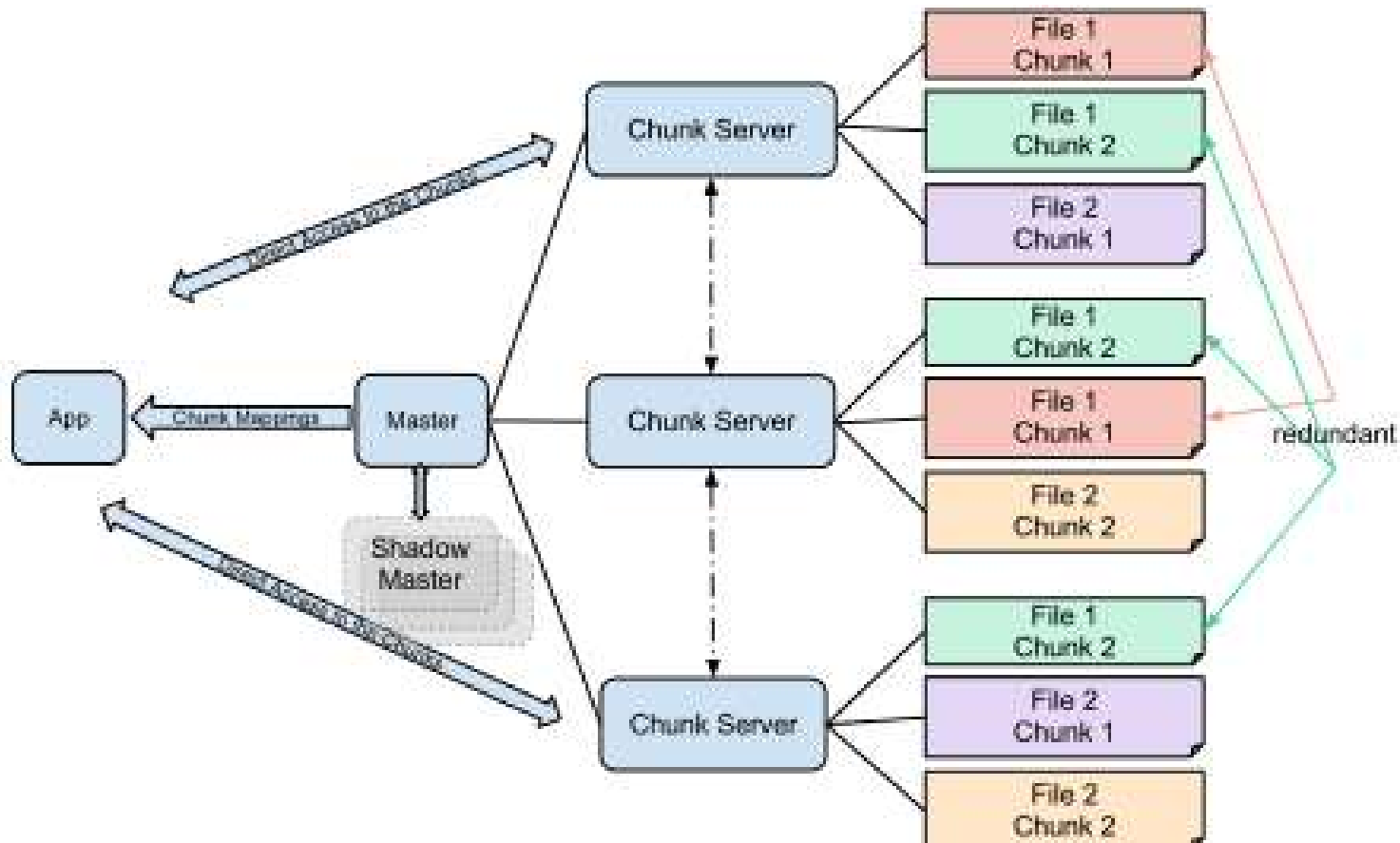
- **HDFS is a multi-node system**
 - **Name Node (Master)**
 - **Single point of failure**
 - **Data Nodes (Slaves)**
 - **Failure tolerant (Data replication)**



- ▶ **Common Amazon Cloud Platform in VM**
 - ▶ HBase & Hive are applications
 - ▶ Other programs or applications may use Hive or HBase as an intermediate layer on the HDFS



Google Big Table System



Google's AppEngine vs Amazon's EC2

Python

BigTable

Other API's



VMs

Flat File Storage



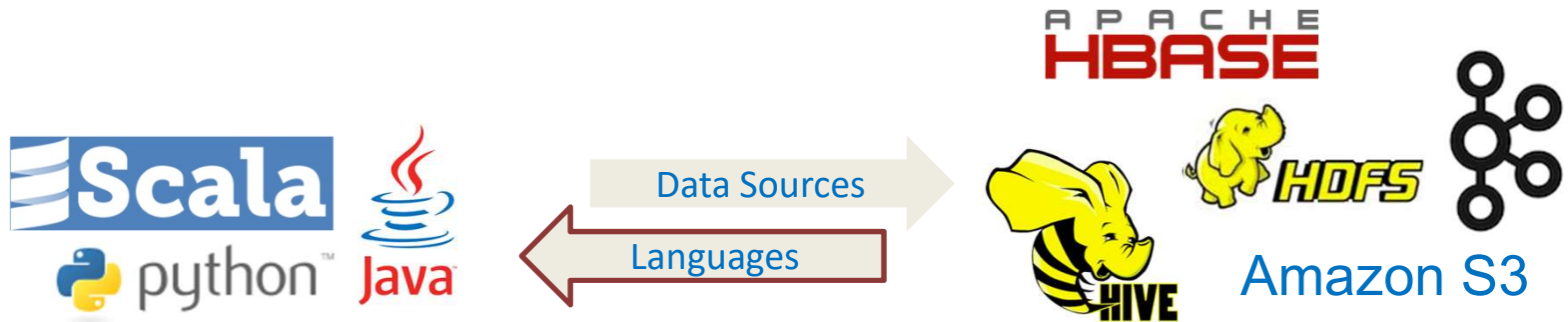
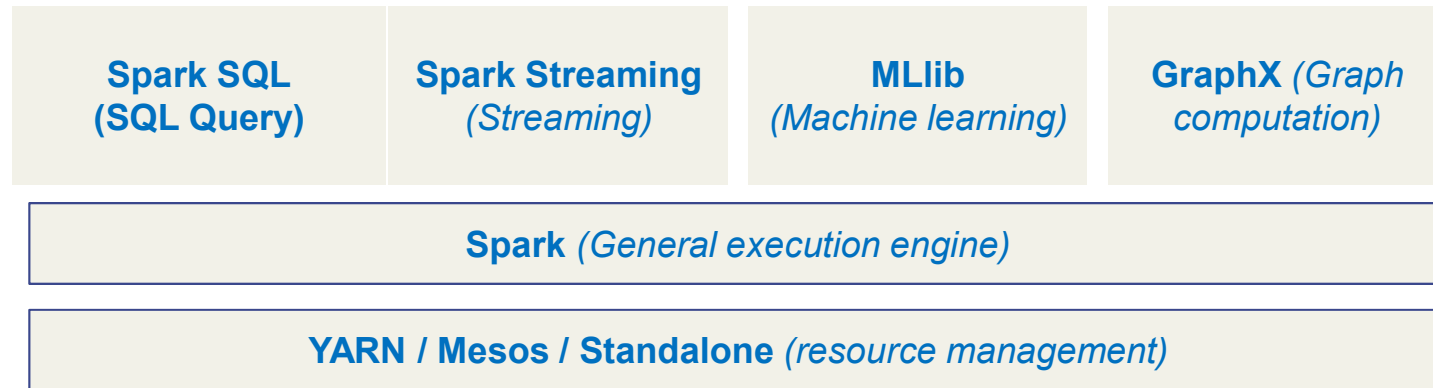
AppEngine:

- Higher-level functionality (e.g., automatic scaling)
- More restrictive (e.g., respond to URL only)
- Proprietary lock-in

EC2/S3:

- Lower-level functionality
- More flexible
- Coarser billing model

SPARK: Real Time Analytic Platform on Cloud

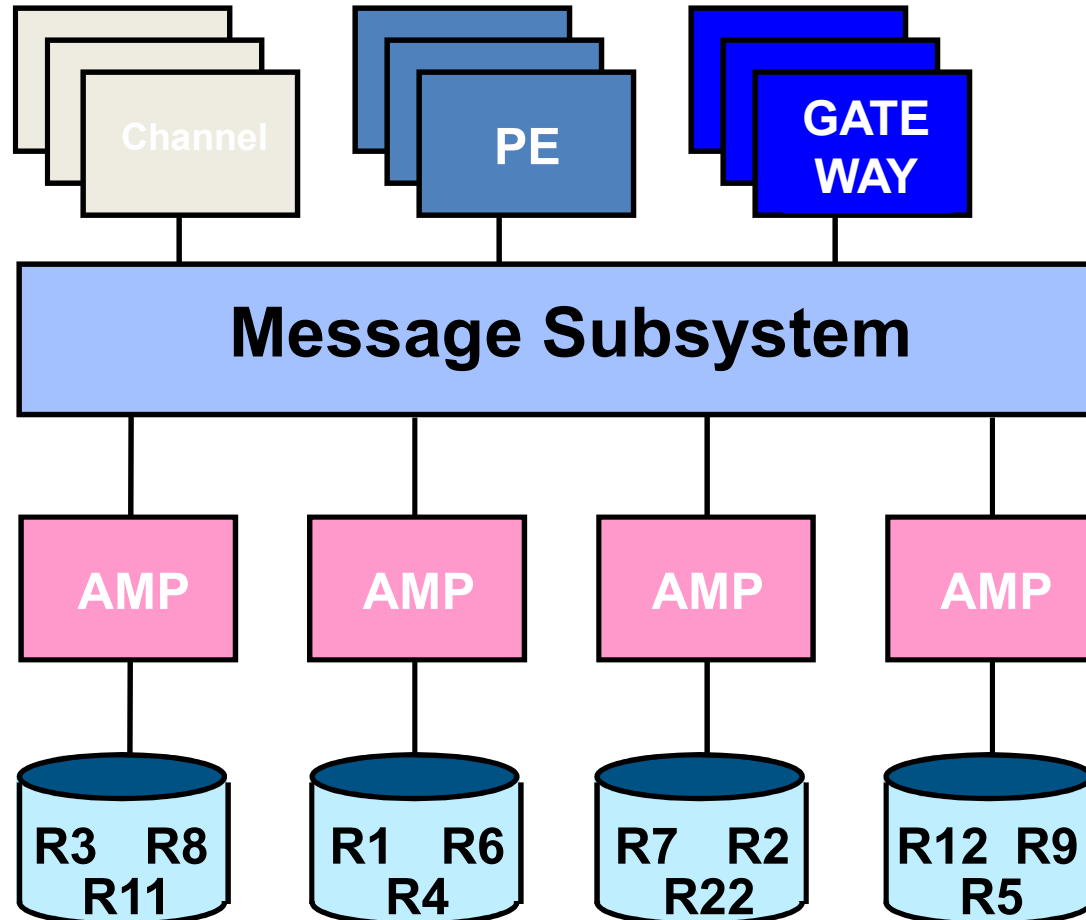


Azure Cloud Platform for Big Data

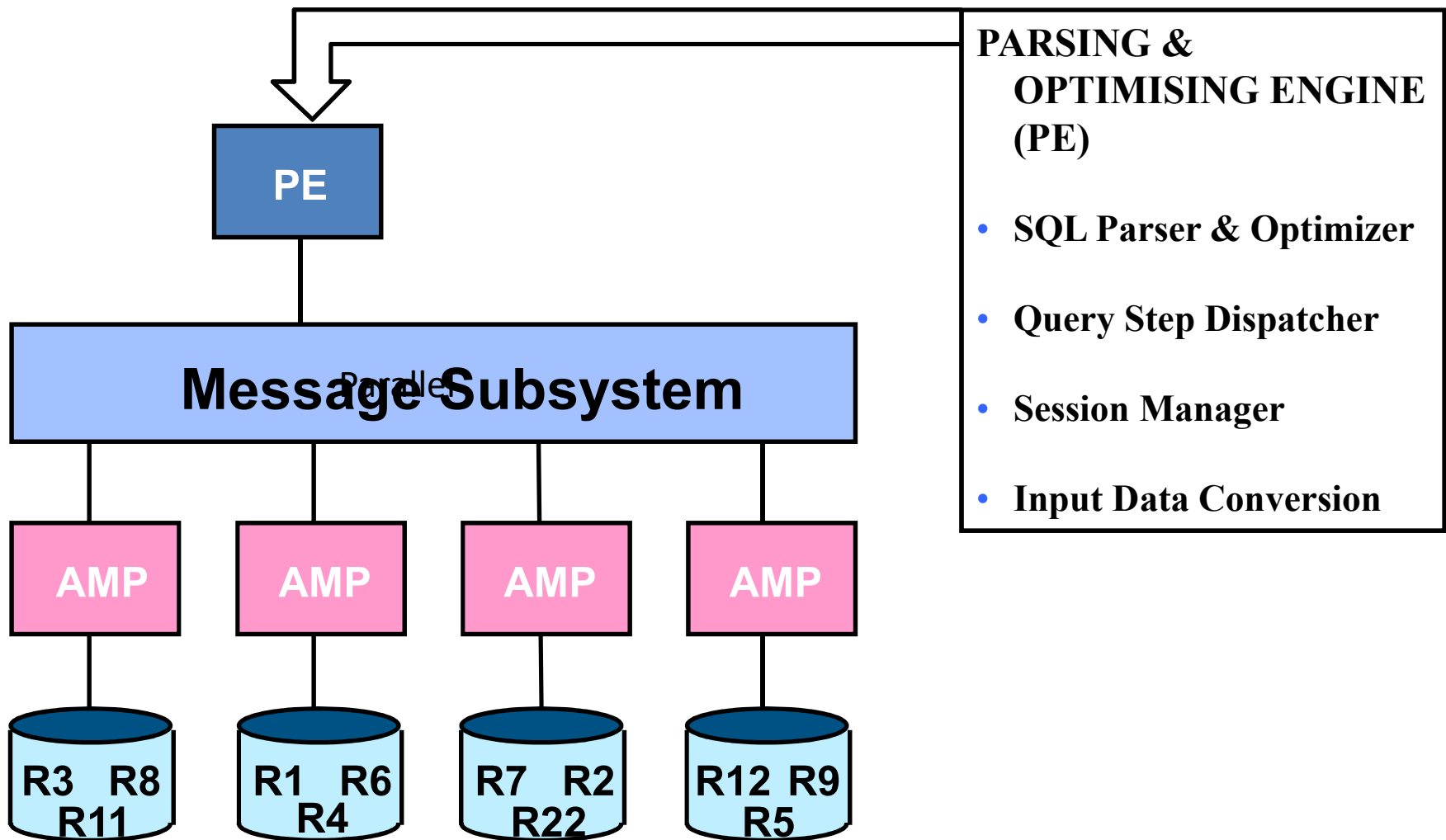
Your Applications



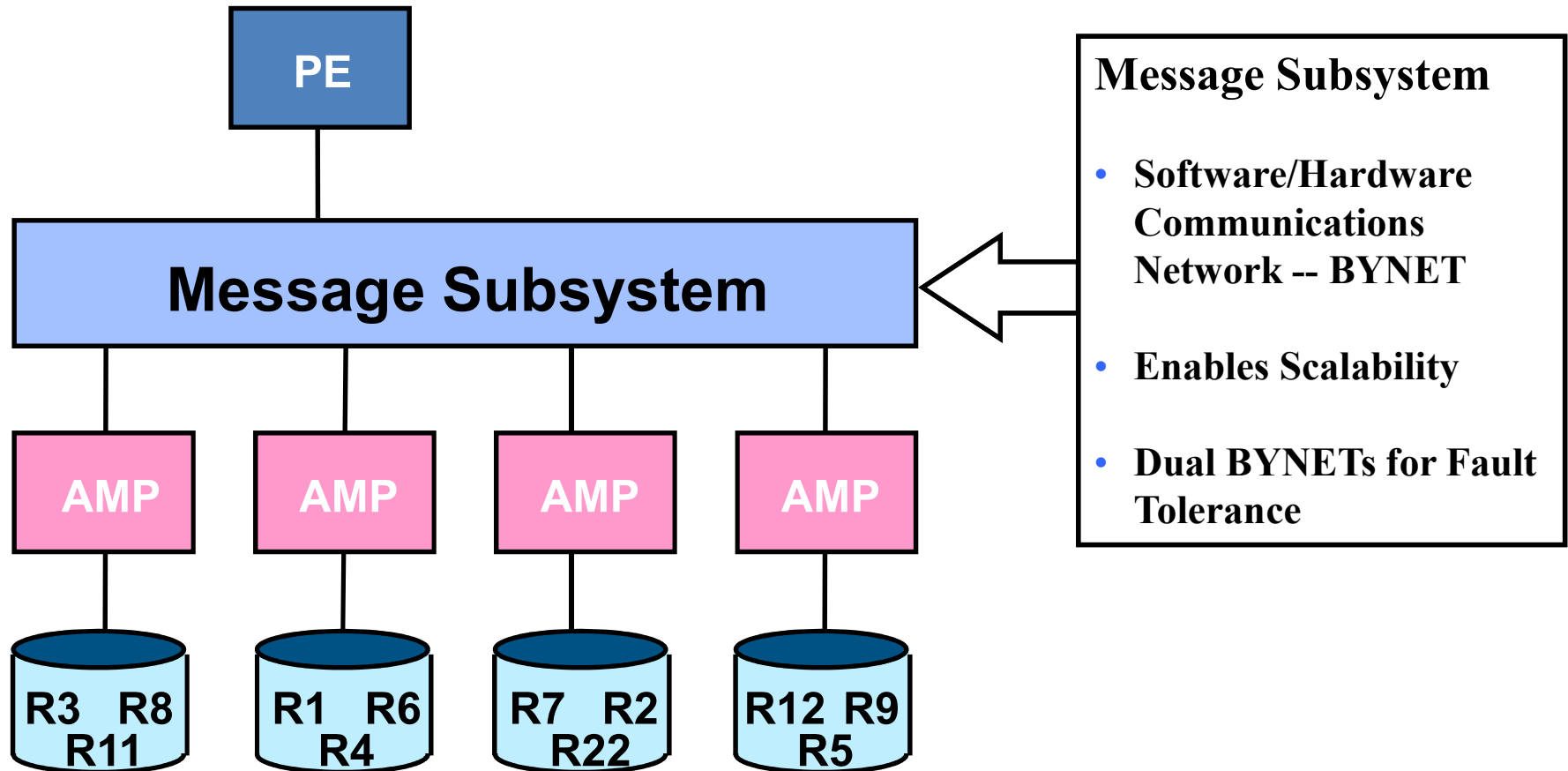
AZURE Parallel Data Warehouse Architecture



Parallel Data Warehouse Architecture



Parallel Data Warehouse Architecture



Parallel Data Warehouse Architecture

